

Replacing Phase 1 with Auction-Dynamics MBO

Approach B, measured against the validated deliverables — and a sixth occasion on which an instrument that could not see an effect reported its absence

August 2026

Abstract

Phase 1 (Γ -convergence relaxation by projected gradient descent) costs 22 h–79 h for a completed $N=300$ solution, and its winner-take-all readout is not a valid partition at that scale, so it is repaired afterwards. This report measures **approach B** — volume-constrained multiphase threshold dynamics [15, 7], with the volume constraint imposed by balanced assignment in the manner of auction dynamics [11] — as an outright *replacement* for Phase 1. Four results. (1) At the pinned Phase 2 campaign, B reaches perimeter **184.4118** and **184.1615** at $N=100$ (two seeds) against the control’s 185.2546, and **319.9428** at $N=300$ against 323.3192 — from 214–248 s of Phase 1 against 48,132 s and 79,069 s, a 211–319 $\times$ reduction in the replaced stage. (2) B produces **0 fragmented cells** at both N on its raw labels, where raw PGD at $N=300$ produces 2 and requires the balanced readout; connectivity is the only one of the three validity gates that has content for B, and it is the one B could have failed. (3) The win is attributable to the MBO dynamics and not to the balanced geodesic initialisation B starts from — which is approach C’s first step — by +2.761% against a pre-registered 0.3% threshold. (4) Of thirteen predictions committed before any run, nine held, three missed and one was initially mis-scored. (5) At $N=400$, where no PGD anchor exists *or is possible*, B yields valid, connected, equal-area partitions at $N=400$, 500 and 750 from 920–3,614 s, agreeing with $\sqrt{N}$ scaling on a fixed mesh to better than 0.06% across three points; at $N=750$ the coarsest ladder rung falls *below* the freeze threshold and has to be dropped, which also removes the level-0 fragmentation seen at $N=300$ and $N=500$ — while on the *same* mesh the incumbent needed 206,344 s at $N=300$ and still failed two of the three gates; the absent baseline is the finding, not a gap. The methodological content is again larger than the technical: a **three-part over-merge instrument was built and detected nothing**, and a per-level claim entered the record having never been measured — the sixth instance of this project’s signature artefact.

Provenance

Status	measured
Date	2026-08-19 (scored results); 2026-08-20 ($N=400$, exploratory); 2026-08-22 ($N=400$ re-verification and export)
Surface	torus $T(R=1.0, r=0.6)$ throughout
Branch	feat/phase1-mbo-auction-dynamics
Code	src/partition/mbo_auction.py (the arm); scripts/run_mbo_arm.py (driver); scripts/score_mbo_arm.py (mechanical scorer); testing/test_mbo_auction.py (G1–G8, NC1–NC5b)
Scored runs	results/arm_mbo_20260819_173411_npart100_V114144_seed84172851, results/arm_mbo_20260819_185131_npart100_V114144_seed61803399, results/arm_mbo_20260819_192659_npart300_V47488_seed61803399, results/arm_init_20260819_181203_npart100_V114144_seed84172851
Exploratory runs	results/arm_mbo_20260820_133709_npart400_V114144_seed84172851 results/arm_mbo_20260822_120646_npart500_V114144_seed84172851 and results/arm_mbo_20260822_182903_npart750_V158260_seed84172851 — $N=400$, 500 and 750, no anchor , and none possible (§10). Scored against nothing; reported for gates, cost and absolute perimeter. The $N=500$ and $N=750$ Phase 2 campaigns were run with <code>--profile</code> . $N=750$ uses a <i>different</i> ladder ($162 \times 154 \rightarrow 410 \times 386$, $V=158,260$) and is therefore outside the fixed-mesh $\sqrt{N}$ comparison. $N=500$ predates the arm experiment.yaml snapshot, so it carries none; its config is parameters/torus_500part_mbo.yaml as of commit e8b8b60, which added an mbo: block after the run at the values the run used.
Anchors	$N=100$: run_20260709_081548, Phase 2 best 185.2546144718457 (iterate 20 of 20), raw PGD, Phase 1 48132.12s. $N=300$: run_20260806_123326, Phase 2 best 323.319247087254 (iterate 18 of 19), PGD <i>plus</i> balanced readout, Phase 1 79068.56s. Both from timing_profile.yaml total_wall_s, never run_time_seconds.
Pre-registration	commit 9ffdefb, 2026-08-19 16:53, <i>before</i> the first scored run at 17:34:11. Results commits b414b67, c8aea83, 5bdc947, 54b937c; corrections d4dc66d, d4b75d7.
Figures	make_figures.py from the committed data.yaml, itself re-derivable by extract_data.py.
Versions	Python 3.12.13, numpy 2.4.4, scipy 1.17.1, h5py 3.16.0; macOS 15.3.1 arm64 (Mac mini).
Anchor check	the harness reproduces the $N=100$ control’s own perimeter <i>bitwise</i> (report 07, 0d); no Phase 2 code has changed since.

1 What B is, and what is not claimed for it

Threshold dynamics originates with Merriman, Bence and Osher [15]: alternate short-time diffusion of the phase indicators with a pointwise reassignment, and interfaces move by approximate mean curvature. Esedoğlu and Otto [7] placed the multiphase case on a variational footing with a Lyapunov functional that Γ -converges to a weighted perimeter — the same limit this project’s own energy targets [3, 17, 16]. Jacobs, Merkurjev and Esedoğlu [11] impose *exact volume constraints* by replacing the pointwise threshold with a balanced assignment solved by Bertsekas’ auction algorithm [2]. **None of this is invented here.**

Per mesh level, the arm iterates on a *hard* labelling $\omega : V \rightarrow \{0, \dots, N-1\}$:

1. **diffuse**: solve $(M + \tau K) y_k = M \chi_k$ for each cell, with one prefactorised LU per level and N back-substitutions;

2. **balanced threshold:** $\omega(i) = \arg \max_k [y_{ik} + \psi_k]$, with ψ from the shared semi-discrete optimal-transport dual [1, 14, 12] that approach A already ships.

There is never a continuous field whose winner-take-all readout can diverge from what was constrained. B therefore does not *close* the continuous–discrete gap that motivated approaches A–E; the category is absent.

The regime, and the novelty claim we do *not* make. Every published implementation of this machinery we are aware of convolves on a *uniform Cartesian grid via FFT*, on a flat domain, at small N , and for Pólya’s *outer* problem — which convex body maximises the length of the shortest partition. That is true of Hu, Liu and Wang [10], whose auction-plus-thresholding machinery is closest to ours but whose domains are $[0, 1]^2$ and 3D boxes on $128^2/256^2$ grids, and of Bogosel and Oudet [4], whose capacity-constrained Voronoi construction with > 100 cells is planar. Ours is the *inner* problem: partition a fixed closed genus-1 surface into N equal-area minimal-perimeter cells on an unstructured triangulated mesh, at N in the hundreds. **We phrase this as “we are not aware of”, never as “first”:** two web searches and two papers is not a literature review, and surface-remeshing and computer-graphics venues may cover adjacent ground under different vocabulary. Stated as a population claim, the corpus is: two full-text readings plus this project’s standing bibliography.

The difference is load-bearing rather than cosmetic. The FEM operator $(M + \tau K)^{-1}M$ is a different object from an FFT convolution, with different failure modes; mesh pinning [18] is ours to manage, and no grid-derived τ heuristic is imported.

2 Result 1 — perimeter

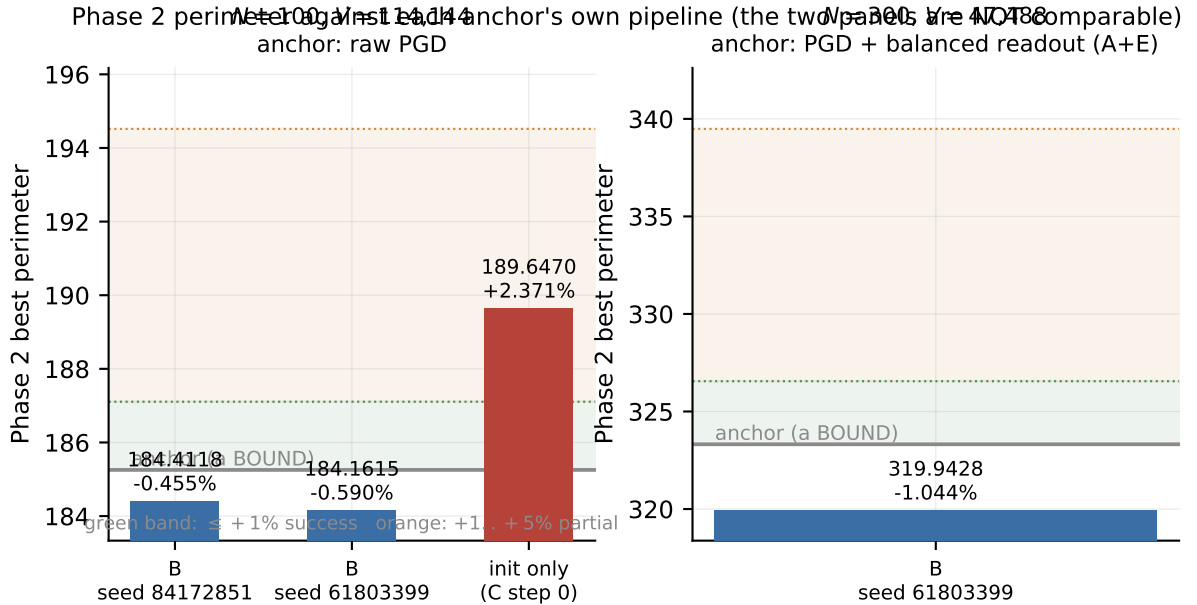


Figure 1: Phase 2 best perimeter against each anchor’s own pipeline. **The two panels are not comparable and must never be pooled:** the $N=100$ anchor is raw PGD, the $N=300$ anchor is PGD *plus* the balanced readout, because raw PGD is not a valid partition at $N=300$.

run	Phase 1	speedup	area worst	frag.	Phase 2 best	vs anchor
$N=100$ s84172851	228 s	210.9×	0.1504%	0	184.4117703525215 (18/20)	− 0.455%
$N=100$ s61803399	214 s	225.2×	0.1493%	0	184.1614647856558 (19/20)	− 0.590%
$N=300$ s61803399	248 s	318.8×	1.4172%	0	319.9427628182375 (16/19)	− 1.044%
init only (C step 0)	67 s	—	0.1355%	0	189.6469917457051 (20/20)	+2.37%

Seed spread at $N=100$ is **0.136%**, which is $3.3\times$ smaller than the margin by which the *worse* of the two seeds beats the control. This is what makes the result scoreable at all: the pre-registration committed that a spread above 1% would mean the primary falsifier could not be evaluated at $\pm 1\%$ and that a single-seed number would be conflating method difference with seed noise.

Six constraints on how these numbers may be stated.

1. **Never pool the two anchors.** They are different baselines. “B beats the best available PGD pipeline at each N , at equal Phase 2 budget” is valid; “B beats PGD by 0.455–1.044%” as a trend in N is not.
2. **The speedup is for the replaced stage only.** End-to-end, including the shared 30–40 min Phase 2, it is $\approx 23\times$ ($N=100$) and $\approx 31\times$ ($N=300$). Against the structure-trigger PGD variant (34,967 s), the incumbent-best figure is $\approx 153\times$.
3. **Neither trajectory is a converged optimum.** Both are at the documented migration-cycling plateau. An earlier draft said “B converged where the control had not”; the sixth review resumed the control’s own Phase 2 for **20 further iterations and found no improvement** (best over all 40 remains 185.2546 at iterate 20; the extension drifts up to ≈ 185.30). B’s plateau is simply 0.455% lower. *That extension is the review’s measurement and has not been independently reproduced by us.*
4. **PGD’s own seed variance is unmeasured.** Every $N=100$ PGD run across both working directories is seed 84172851, and every $N=300$ $\lambda=11.5$ run is seed 61803399. Each anchor is a single trajectory. “B beats PGD by more than PGD’s own seed lottery” is *not* established.
5. **The $N=300$ margin cannot be decomposed, even in principle.** The readout costs +1.882% in label-boundary length (189.7310 raw $\rightarrow$ 190.7114 shifted $\rightarrow$ 193.3024 repaired, the last being what the anchor’s Phase 2 consumed). B’s raw boundary is 187.2574: shorter than the repaired labels by 3.127% *and* shorter than raw PGD’s own invalid labels by 1.304% — the strong form, since imbalance can only make a boundary artificially short. So the margin is not merely “not paying the repair bill”. But how much is un-erased repair residue cannot be separated, because **no valid repair-free PGD partition exists at $N=300$.**
6. $N=300$ rests on one Phase-2-scored seed.

3 Result 2 — validity, and which gate actually has content

“All three gates pass” is true and misleading

An earlier write-up of this result — ours — said B’s raw labels pass all three validity gates. Phase 0’s own findings say why that is the wrong emphasis:

- **Dormant is vacuous** for any balanced-assignment arm. One-hot labels give peak density 1.0 by construction; the gate cannot fail. The harness already emitted `vacuous_for_arms: True`.
- **Area is near-expected**. Equal area is precisely what B’s assignment step *optimises*. Passing means the solver converged; Phase 0 defined it as the *solver-failure detector*, not a validity test.
- **Connectivity is the only gate genuinely at risk**. Nothing in B enforces it, and the source proposal states it is “not per-step guaranteed”.

The defensible claim is narrower and still strong: **B produces 0 fragmented cells at $N=300$, where raw PGD produces 2** (with 10 imbalanced cells, worst 36.15%) — which is why PGD’s baseline required the balanced readout and B required nothing.

Re-verified through `testing/check_fragmentation.py`, which shares no code with the driver’s reporting: all four arm solutions pass, with **0 cells having more than one raw component including sub-threshold speckle**.

Robustness. Three further $N=300$ MBO ladders at seeds 27182818, 13131313 and 84172851 give **0 fragmented** in every case, with worst area 1.2531/1.4467/1.3887% and label boundaries 186.08/186.55/186.90 — all *shorter* than the scored seed’s 187.2574. Ladder only, no Phase 2; *review-produced, not independently reproduced by us*.

4 Result 3 — attribution: is this B, or is it C’s first step?

B is initialised from one balanced C-scores assignment — farthest-point seeds, multi-source Dijkstra, scores $-d^2$, balanced assignment — which is literally approach C’s step 0 [13, 1, 6]. If MBO barely improved on it, the number in §2 would be a result about C reported under B’s name, and *nothing* in the gate suite would have detected that.

	Phase 1	label boundary	Phase 2 best	censored
init only (C step 0)	67 s	116.1989	189.6469917457051 (20/20)	yes
B (MBO)	228 s	107.6014	184.4117703525215 (18/20)	no
control (PGD)	48,132 s	107.3410	185.2546144718457 (20/20)	plateaued

$$\text{P13} = \frac{189.646992 - 184.411770}{189.646992} = +\mathbf{2.761\%} \quad \text{against a pre-registered threshold of 0.3\%}.$$

Two disclosures. The measured 2.761% is an **upper bound** on MBO’s contribution, because the init arm is censored: its converged value is ≤ 189.6470 , and a lower true value shrinks the gap. For attribution to fail the init would have to reach ≈ 185.0 — another $\approx 2.5\%$ on a trajectory gaining ≈ 0.038 per iterate and decelerating. And the init-only control was run on the *finest* mesh, i.e. the *strongest* member of the init family, which biases the test *against* B.

An unflattering free datum on C. C’s step 0 plus Phase 2 lands at +2.37% above the PGD control — a PARTIAL verdict under this project’s own bands — for 67 s of Phase 1. Real C iterates Lloyd and would improve on this, so it is a *lower bound* on C; but C’s starting point is behind

both B and PGD, which is directly relevant to whether C is worth building. Note also that C minimises a *quantisation* energy, not perimeter; the two optima coincide only asymptotically and only in the plane [9, 8], and this torus has Gaussian curvature of both signs.

5 The mesh ladder, and τ

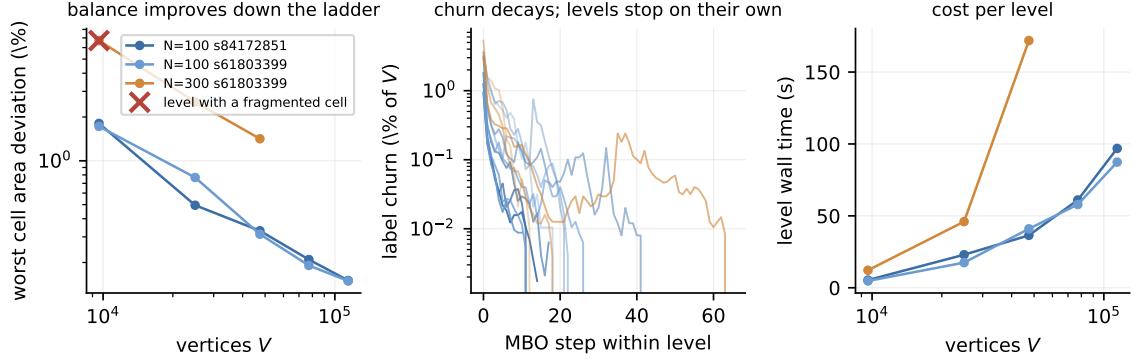


Figure 2: Per-level behaviour. Left: worst cell area deviation improves monotonically down the ladder; the cross marks the one level carrying a fragmented cell. Middle: label churn decays within every level, so levels terminate on their own rather than at the iteration cap. Right: cost per level.

$\tau = \min((c h_{\text{mean}})^2, (\rho R_{\text{cell}})^2)$ with $c = 4$ and $\rho = 1.0$, pinned before any run. $c = 4$ rather than $c = 2$: the latter is the freeze regime [18]. The cap is the over-merge side of a two-sided window, and it binds on exactly one level in use, $N=300$ level 0.

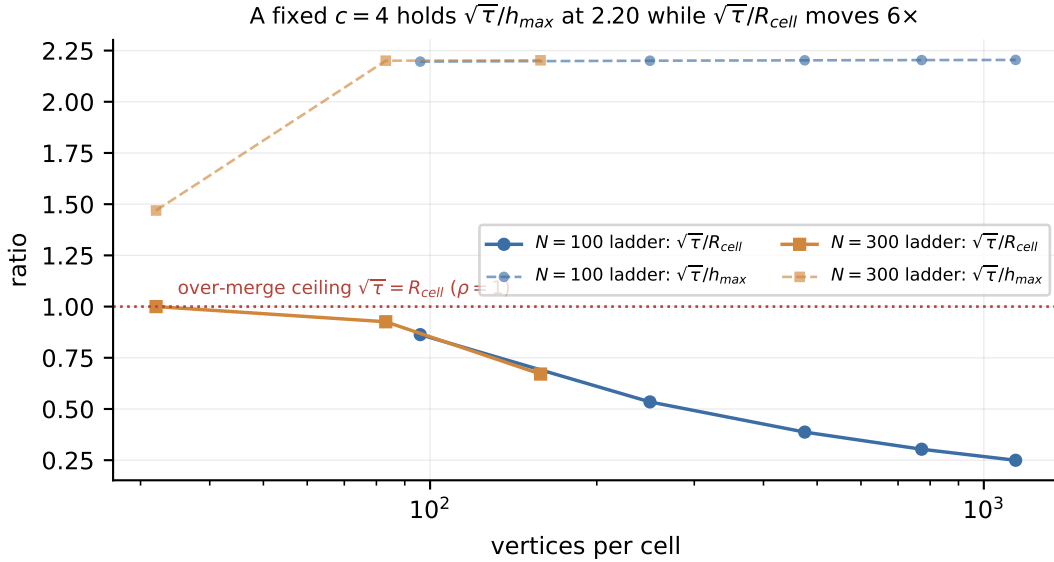


Figure 3: A fixed c does *not* hold the window. $\sqrt{\tau}/h_{\text{max}}$ stays at 2.20 across every level of both ladders while $\sqrt{\tau}/R_{\text{cell}}$ moves $6\times$. Reporting on h_{mean} alone flattens the coarse band: this mesh is $1.81\times$ anisotropic.

6 The headline null: an instrument that detected nothing

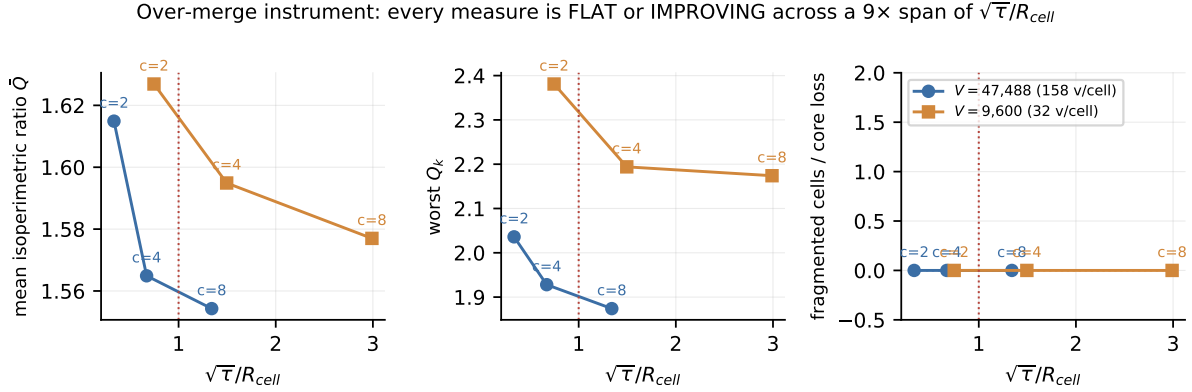


Figure 4: Fragmentation, per-cell isoperimetric ratio $Q_k = P_k^2/(4\pi A_k)$, and core loss, swept over $c \in \{2, 4, 8\}$ at two resolutions. Every measure is flat or *improving*; none registers an over-merge penalty anywhere, including at $\sqrt{\tau}/R_{\text{cell}} = 2.99$, three times the pinned cap.

Assignment quality *cannot* see over-merging — the balanced assignment hits its area target by construction, so a smeared, non-local cell scores as well as a compact one. (An earlier draft of the plan cited a healthy assignment-quality fixture as evidence against over-merging; that was a measurement of a different quantity than the claim it supported, and it was withdrawn.) So three geometry instruments were built. Across a $9\times$ span of $\sqrt{\tau}/R_{\text{cell}}$ and two levels differing $5\times$ in resolution, **not one of them registered anything**: core loss and fragmentation are 0 in all six configurations, and compactness moves monotonically the *other* way.

The instruments are demonstrably *responsive* — $\bar{Q}$ moves 3.8%, Q_{worst} 8%, boundary length 1.9% — they simply move toward improvement. What is established is therefore that **no damage is detectable**, *not* that over-merging cannot occur.

Standing limit

The ρ question is **open**, and no future claim about c or ρ may lean on this report’s evidence in either direction. The cap was retained because it is pre-registered and because it biases *against* B — it lowers c to 2.68 at $N=300$ level 0, and $c=2$ measures worse than $c=4$. That is a discipline argument, not an empirical one. **Named follow-up**: $c=8$ showed no penalty on any instrument *and* a 0.33% shorter starting boundary — a third of the entire +1% success bar — so the cap may be costing perimeter for no measured benefit. That is a separate arm, scored through the same harness.

7 Why there is no exactly-monotone energy, and what is gated instead

Esedoglu–Otto monotonicity is a minorise–maximise argument requiring (i) the same symmetric positive semi-definite form in the energy and in the assignment objective, and (ii) an *exact* assignment. **Both fail here, for independent reasons.** The theorem’s object is $MA_\tau = M(M + \tau K)^{-1}M$, symmetric PSD; our step maximises $\langle \chi', Dy \rangle$ with $D = \text{diag}(v)$ *lumped*, and DA_τ is not symmetric. And Jacobs, Merkurjev and Esedoglu obtain a theorem because Bertsekas’ auction is exact, whereas `solve_dual_offsets` is subgradient ascent returning its best iterate. **This is the price of a substitution made deliberately in Phase 0** — where Sinkhorn [5]

and auction were both measured and both lost to the incumbent on assignment quality — paid here rather than there.

What survives needs neither concavity nor exactness. Since $\omega'(i) = \arg \max_k [y_{ik} + \psi_k]$, multiplying the pointwise optimality by $v_i > 0$ and summing gives

$$\langle \chi', Dy \rangle - \langle \chi, Dy \rangle \geq \sum_k \psi_k (T_k - T'_k)$$

whose right-hand side is exactly the slack that inexactness costs; it vanishes at perfect balance. Measured slack runs 1.7×10^{-4} to 5.5×10^{-3} — five to seven orders of magnitude above the 10^{-9} tolerance an earlier draft proposed for energy monotonicity, so that tolerance was never defensible. **Real E_τ increases do occur** (4 of 96 active steps at $N=300$), and gate G4 therefore gates the inequality, whose measured violation is $\leq 6.13 \times 10^{-8}$, while E_τ is reported as a descriptive trend only.

8 Predictions, and what they caught

	PREDICTED (committed 9ffdefb)	MEASURED	
P1	188.0 (+1.5%), PARTIAL	184.4118 / 184.1615, SUCCESS	MISS
P2	spread < 0.5%	0.136%	HELD
P3	<= 0.28%, point 0.15%	0.1504 / 0.1493%	HELD
P4	<= 3, point 0	0	HELD
P5	>= 30x, point 53x	210.9 / 225.2x	HELD (point missed)
P6	331.4, 80% CI [323.3, 341.0]	319.9428	MISS
P7	<= 2.02%, <= 6 frag, >= 88x	1.4172%, 0, 318.8x	HELD
P8	<= 2 of 5 flagged	0, clauses agree	HELD
P8'	dE flags L2,L3,L4 (falsifying P8)	dE flat 14-22, flags none	MISS
P9	frag not death; worst at L0	1 frag at L0 (295), 0 elsewhere	HIT
P10	viol <= 1e-12; excursions <= ceiling	-6.13e-08; 4/96 within ceiling	HELD
P12	no abort	20/20, 19/19	HELD
P13	>= 0.3%	+2.761%	HELD

Figure 5: Thirteen predictions committed in 9ffdefb at 16:53, before the first scored run at 17:34:11.

P1 and P6 missed, and P1's reasoning is refuted. The rationale was that B sees perimeter only through a coarse- τ surrogate while PGD saw the actual energy for 13.4 h. B also *enters* Phase 2 from a 0.243% *longer* boundary than the control and still finishes ahead — a genuine crossover, so the advantage is not initial geometry. That it is combinatorial is a hypothesis consistent with these data, not established by them. P6 missed worse: its 80% interval's lower bound was the anchor itself, so essentially no probability was assigned to B beating PGD + A + E at $N=300$.

P8' missed and invalidated its own model. It predicted, from a log-linear interpolation of $\Delta E/\text{tail}$ in the freeze ratio c/c_{lo} , that levels 2–4 would be flagged pinned. Measured, $\Delta E/\text{tail}$ is

flat across the ladder — 15.76, 21.62, 15.11, 14.22, 16.49 against a threshold of 79.25 — and level 4, sitting exactly at $c/c_{lo} = 0.999$, scores 16.49 where the model demanded ≈ 260 . The freeze ratio does not predict pinning here.

9 The sixth artefact — and it is ours

Report 07 catalogued five occasions on which a measurement artefact was read as a result about the method. This is the sixth, and it was produced in this work.

The plan asserted **“0 fragmented at every level”** and, on that basis, scored P9 — which predicted over-merge would appear as fragmentation rather than cell death, worst at level 0 — as UNTESTABLE, “the predicted phenomenon never occurred”. **Until the sixth review, LevelReport carried no connectivity field at all.** Per-level fragmentation had never been computed. The claim was an inference presented as a result, and it was false.

With the instrument added and the $N=300$ ladder re-run to a bit-exact reproduction (worst areas 6.7063/2.5394/1.4172%, steps 16/21/66, boundary 187.2574):

level 0: 1 fragmented cell (cell 295). Levels 1 and 2: 0. No cell death.

The phenomenon occurred, at exactly the level P9 named, in exactly the form P9 predicted. **P9 is a hit, not untestable** — “untestable” had absorbed a correct prediction on the strength of a measurement never taken. The stray is healed by level 1’s MBO run, which is direct evidence for the mechanism the taxonomy predicted: stray islands are perimeter-inefficient and shrink under curvature flow, the opposite dynamic to the discrete-area trim that manufactured them.

The fix is instrumentation — `run_gates` per level, seconds of cost, now committed — not a resolution to be careful.

10 Result 5 — $N=400, 500, 750$: existence where the anchor cannot exist

The three results above are all *margins*: B against a control, at a pinned Phase 2 budget. That design has a ceiling built into it. It can only be run where a control exists, and the control is the method whose cost is the reason B was built. Pushed far enough in N , the comparison stops being expensive and starts being impossible.

$N=400, 500$ and 750 are past that point, and this section reports them as what they are: **existence-and-validity** results with no anchor, no margin, and no comparison of perimeters. All three ran in the driver’s exploratory `--config` mode, where the mesh-match assertion is skipped because there is nothing to match. $N=500$ deliberately reuses $N=400$ ’s ladder unchanged, which converts the $\sqrt{N}$ argument below from a single check into a three-point one; $N=750$ could not, for a reason that is itself a result (§10).

The results

$N=400$ and $N=500$ run to $V = 114,144$; $N=750$ runs to $V = 158,260$. All three validity gates pass on the *raw* labels at every N , re-verified with `testing/check_fragmentation.py` independently of `arm.report.yaml`:

	$N=400$	$N=500$	$N=750$
Phase 1 wall	920 s	1,322 s	3,614 s
dormant	0 / 0	0 / 0	0 / 0
area imbalance (bar)	0, 0.7773% (1.1214)	0, 1.0993% (1.4017)	0, 1.2569% (1.5165)
connectivity	0	0	0 at <i>every</i> level
Phase 2 perimeter	368.6603 (it. 19/20)	412.1138 (it. 10/20)	504.9214 (it. 14/20)
variable points	29,287	32,704	47,088
max area violation	8.18e-7	1.10e-6	4.68e-7

All three Phase 2 constraint residuals are converged to tolerance, so the partitions are genuinely equal-area rather than approximately so, and the $N=500$ and $N=750$ bests are *mid-trajectory* — not censored; both descend and then oscillate in the migration-cycling plateau. All three deliverables are exported with `finalised=True`.

The $\sqrt{N}$ cross-check, three points

Not a gate, and the sharpest instrument available in the absence of an anchor. On a *fixed* mesh total perimeter should scale as $\sqrt{N}$; all three runs below sit on the same $V = 114,144$ mesh, so the comparison is free of any refinement effect. $N=500$ was given $N=400$'s ladder unchanged precisely to make this test available, and its predicted value was recorded in `parameters/torus_500part_mbo.yaml` *before* the run.

	predicted	measured	deviation
$N=100 \rightarrow N=400$	368.8236	368.6603	-0.044%
$N=400 \rightarrow N=500$	412.1748	412.1138	-0.015%
$N=100 \rightarrow N=500$	412.3573	412.1138	-0.059%

Three independent runs at three values of N agree with the scaling to better than 0.06%. All three deviations are negative, which is a consistent slight shortfall rather than scatter; with three points we do not attempt to say why.

$N=750$ is *not* in that table, because it runs on a different mesh and the test is only sharp at fixed V . Recorded for completeness and scored against nothing: the fixed-mesh value implied by $N=500$ is 504.7342 and the measurement is 504.9214, +0.037%. Two points worth stating plainly. The magnitude is small, but the *sign* is opposite to all three fixed-mesh checks, and the pre-registered expectation in `parameters/torus_750part_mbo.yaml` was that a finer mesh should land slightly *below* — so the stated direction was wrong. Nothing is concluded from a single unscored point; it is noted so the error is on the record.

The ladder had to change at $N=750$, and the diagnostics said so first

Every earlier config began at 100×96 . At $N=750$ that rung is unusable, and this is visible *before* running: it gives 12.8 verts/cell and $\sqrt{\tau}/h_{\max} = 0.929$ against a local non-freeze threshold of 0.964 — a ratio of **0.964**, **below 1**, so the coarse-triangle band cannot move at all. Every measured run had been above 1 at every level ($N=400$ min 1.049, $N=500$ min 1.067).

The mechanism is a squeeze that tightens with N : more cells means smaller cells, the over-merge cap holds $\sqrt{\tau} \leq R_{\text{cell}}$, so the diffusion length shrinks — while the coarse mesh's triangles do not. Past some N the diffusion no longer reaches across one triangle and the level freezes. **The remedy is the ladder, not τ** : raising τ_c walks toward the over-merge wall on the other side of the same two-sided window. The config therefore keeps levels 1–5 verbatim and drops the dead rung, running $162 \times 154 \rightarrow 410 \times 386$. Every level then sits at ratio ≥ 1.131 , and the finest level's margin (1.131) is better than $N=500$'s (1.109) and $N=400$'s (1.049), both of which ran clean.

It worked, and it bought something beyond avoiding the freeze: $N=750$ is the first high- N run with **zero fragmented cells at every level**, where $N=300$ and $N=500$ each produced one at level 0 that later levels had to heal. Starting at 33 verts/cell instead of 12.8 or 19.2 removes the defect rather than repairing it.

Two mechanisms recur at a third N

Neither is new; both were predicted or observed earlier, and $N=500$ is an independent occasion for them to fail.

Level-0 fragmentation, healed by level 1. $N=500$ level 0 produces one fragmented cell (323) and level 1 has none — exactly the pattern §9 recorded at $N=300$ (cell 295), and the mechanism the taxonomy predicts: stray islands are perimeter-inefficient and shrink under curvature flow. $N=400$ shows none at any level. This is visible only because the per-level instrument now exists; it is the same claim that was once asserted with nothing measuring it.

The over-merge null holds. At $N=500$ the cap binds on two levels and $\sqrt{\tau}/R_{\text{cell}}$ is at its ceiling there, yet the instrument reports core loss = 0 at every level, q_{worst} falling $2.95 \rightarrow 1.81$ down the ladder, and q_{mean} in 1.54–1.61 against $N=400$ ’s 1.56–1.57. A third N , a tighter cap, and the same null of §6. Still undetected rather than excluded.

Why the missing anchor is the finding

The obvious objection is that a result without a baseline is weak. Here it is the reverse, and the reason is measurable rather than rhetorical. Compare at the **matched** finest mesh $V = 114,144$ — the same mesh, so no refinement effect enters:

	PGD, $N=300$ run_20260808_191030	B, $N=400$ arm_mbo_20260820_133709
Phase 1 wall	206,344 s (57.3 h)	920 s
dead / weak	0 / 0	0 / 0
area imbalance	3 imbalanced, worst 24.81%	0, worst 0.78%
fragmented	2	0
valid on raw labels?	no — requires the readout	yes

On the identical mesh, with *one hundred fewer cells*, the incumbent took **224×** the wall time and failed two of the three gates. Requiring a PGD anchor at $N=400$ therefore asks the incumbent to do precisely the thing whose absence motivated B. The honest claim needs no margin:

Approach B produces a valid, equal-area, fully connected 400-cell partition on a 114,144-vertex torus from 15 minutes of Phase 1, in a regime where the incumbent has already ceased to produce valid partitions at $N=300$ on the same mesh.

What this section does *not* claim

The -0.455% and -1.044% margins of §2 are not extended to $N=400$ or $N=500$, and no perimeter here is compared against PGD. The matched-mesh table above is a statement about *validity and cost*, both measured on both sides at the same V ; it is not a perimeter comparison, and the two must not be blurred. The $\sqrt{N}$ check is B against B, so it tests internal consistency, not correctness against an independent method: three runs agreeing on a scaling law they all obey would not detect an error common to all three. And $N=500$ reuses $N=400$ ’s ladder and seed, so the two are not independent in the mesh or the initialisation, only in the run.

The cost centre has moved — and it is not the solver

Phase 2 took 7,326 s, 8,829 s and **18,323 s** (5.09 h) at $N=400/500/750$, against Phase 1’s 920 s, 1,322 s and 3,614 s. The stage this report replaces is therefore 11%, 13% and 16.5% of end-to-end cost — the last rising only because $N=750$ was given an extra ladder level. The 211–319× of §2 was always a statement about the replaced stage; past $N=400$ the remainder dominates.

The $N=500$ and $N=750$ campaigns were run with `--profile`, and the breakdown is not what the forward plan assumed. **The IPOPT solver accounts for 5.2% of the campaign at $N=500$ and 3.6% at $N=750$ — a *shrinking* share of a growing problem.** At both, IPOPT runs 201 solver iterations per topology iteration, hitting `max_opt_iter:` 200 every time, and the internal callback split is essentially identical (Hessian 40.4% vs 40.7% of solver time). So 95–96% of Phase 2 is spent *outside* `optimize()`: contour rebuild, Steiner setup, migration detection and checkpoint export with its roundtrip verification. **We establish the solver’s share, not the composition of the remainder** — decomposing that is the measurement `docs/plans/PUBLICATION_READINESS_PLAN.md` Phase 4 should now make, in place of its original premise that the exact-Hessian path was the target.

Phase 2 also scales superlinearly in variable points, close to quadratically. Campaign wall against variable-point count:

	variable points	wall	s/VP	exponent
$N=400$	29,287	7,326 s	0.250	
$N=500$	32,704	8,829 s	0.270	1.69
$N=750$	47,088	18,323 s	0.389	2.00

Three points cannot fit a law, and the two exponents are not equal, so the defensible statement is *superlinear and steepening*, not “quadratic”. Either way it is the binding constraint on N : at this trend $N=1000$ implies a Phase 2 of roughly 7–9 h, against a Phase 1 that remains under an hour.

A reporting trap in `timing_profile.yaml`

The Phase 2 profile’s `summary.total_wall_s` is 459.27 s while the campaign took 8,829 s. The field accumulates only time inside `optimizer.optimize()`; everything else in the loop is outside it. A **19×** under-report if read as campaign wall — the same class of trap as `run.time.seconds` in the Phase 1 metadata, which under-reports by 3.6× and is already documented. Campaign wall must be taken from the iterate timestamps.

Two scaling consequences worth recording. The τ over-merge cap binds on *two* levels at both N (against one at $N=300$ and none at $N=100$) as cells shrink relative to the coarse mesh. And the vertex-granularity floor has risen from 0.5607% to 0.7009%, pushing the worst cell from 1.39× to 1.57× that floor against a 2× bar — so the finest mesh will have to grow before the bar is reached, and that raises Phase 2, which is now the expensive half.

11 Threats to validity that no instrument here addresses

1. **PGD’s seed variance** (§2, constraint 4). Both anchors are single trajectories; the corpus was enumerated across *both* working directories, since `results/` is gitignored and each worktree has its own.
2. **Over-merging** (§6). Not excluded, merely undetected.
3. **Phase 2 budget as a confound**. B may start from geometry that lets Phase 2 converge *within* a budget PGD cannot use as well. The 40-iteration control extension argues against this, but is review-produced.

4. **Same-machine identity** of the July control and the August arms is consistent with the recorded hardware but is not stamped in run metadata.
5. $N=1000$ is untouched. Nothing here establishes that B’s behaviour extrapolates.

12 Conclusions

1. B replaces Phase 1 at $211\text{--}319\times$ less wall time in the replaced stage and reaches a *shorter* Phase 2 perimeter than the best available PGD pipeline at both N , at equal Phase 2 budget.
2. Its raw labels are connected at both N — the one validity gate B could have failed — so the readout and repair stages are unnecessary for B.
3. The advantage is attributable to the MBO dynamics, not to the balanced geodesic initialization, by $+2.761\%$ against a 0.3% threshold.
4. Three pre-registered predictions missed, all in B’s favour, and one was initially mis-scored in B’s *disfavour* by an instrument that did not exist.

Provenance of the two closest references

Both papers are held at `docs/papers/`. Authors, titles and versions are now **verified against their title pages**: [10] is Shilong Hu, Hao Liu and Dong Wang (v1, 25 May 2024) — an earlier draft of this report guessed two of the three given names wrongly, from a secondary description — and [4] is Benjamin Bogosel and Edouard Oudet (v2, 7 June 2021).

The scope characterisation is now partly checked directly rather than taken on trust: [10] states in its abstract that it treats the *longest* minimal length partition problem by Gaussian convolution with auction dynamics and threshold dynamics, on 128×128 and 256×256 grids in 2D and 3D, and its full text contains **zero** occurrences of “manifold”, “Riemannian” or “torus”; [4] states in its abstract that it seeks the set maximizing the perimeter of the least-perimeter partition and uses capacity-constrained Voronoi diagrams as *initialization*, with 44 occurrences of “convex” and 2 of “manifold”. **What remains second-hand** is the reading of their full numerical sections; the author has read the title pages, abstracts and these keyword counts, not the papers end to end.

Volume and page numbers throughout `docs/math/shared/references.bib` carry an upstream note that they should be checked against published sources before external distribution. That note still stands for the other 28 entries, none of which this report verified.

Related documents

- Plan, pre-registration and full correction history: `docs/plans/PHASE1_BC_REPLACEMENT_PLAN.md`
- Taxonomy: `docs/reference/PHASE1_HIGHN_APPROACHES_ABCDE.md`
- The instrument this report is scored through: `docs/experiments/07-phase0-shared-harness/`
- The gap B makes vacuous: `docs/reference/winner_take_all_partition_gap.md`

References

- [1] Franz Aurenhammer, Friedrich Hoffmann, and Boris Aronov. Minkowski-type theorems and least-squares clustering. *Algorithmica*, 20(1):61–76, 1998.
- [2] Dimitri P. Bertsekas. The auction algorithm: A distributed relaxation method for the assignment problem. *Annals of Operations Research*, 14(1):105–123, 1988.
- [3] Benjamin Bogosel and Édouard Oudet. Partitions of minimal length on manifolds. *Experimental Mathematics*, 26(4):496–508, 2017. THE foundational reference for this project: Phase 1 (Γ -convergence relaxation) and Phase 2 (direct perimeter minimization). Distinct from `bogosel2021longest`, which is the same authors on the OUTER problem and is cited only in the literature section of experiment 08 – do not conflate them. read `year = 2023` with no volume or pages, under the key `bogosel2023partitions`. The rest of the repository already said 2017 (`src/surfaces/double_torus.py`, `banchoff_chmutov.py`, `docs/plans/MESH_DEGENERACY_AND_NEEDLE_TRIANGLES.md`); the bibliography was the outlier. Taylor & Francis lists this as *Experimental Mathematics volume 26 (2017), issue 4*, and the article page gives “Published online: 04 Oct 2016”. Crossref carries both dates and its automatic BibTeX transform emits `year = 2016`; the DOI also encodes 2016. We cite the ISSUE year, 2017. A venue preferring online-first dating would write 2016; both refer to this same article, DOI 10.1080/10586458.2016.1223570, arXiv:1606.02873.
- [4] Benjamin Bogosel and Edouard Oudet. Longest minimal length partitions. *arXiv preprint arXiv:2102.02891v2*, June 2021. Outer problem: numerical evidence that the ball maximizes the perimeter of the least-perimeter partition under volume constraint. Uses capacity-constrained Voronoi diagrams as INITIALIZATION. Verified against `docs/papers/2102.02891v2.pdf` (v2, 7 June 2021); 44 occurrences of “convex”, 2 of “manifold”.
- [5] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.
- [6] Fernando de Goes, Katherine Breeden, Victor Ostromoukhov, and Mathieu Desbrun. Blue noise through optimal transport. *ACM Transactions on Graphics*, 31(6):171:1–171:11, 2012.
- [7] Selim Esedoğlu and Felix Otto. Threshold dynamics for networks with arbitrary surface tensions. *Communications on Pure and Applied Mathematics*, 68(5):808–864, 2015.
- [8] Allen Gersho. Asymptotically optimal block quantization. *IEEE Transactions on Information Theory*, 25(4):373–380, 1979.
- [9] Thomas C. Hales. The honeycomb conjecture. *Discrete & Computational Geometry*, 25(1):1–22, 2001.
- [10] Shilong Hu, Hao Liu, and Dong Wang. Iterative thresholding methods for longest minimal length partitions. *arXiv preprint arXiv:2405.16040v1*, May 2024. Auction dynamics + threshold dynamics for the OUTER (Pólya) problem; Gaussian convolution on 128×128 and 256×256 Cartesian grids, 2D and 3D. Authors verified against the title page of `docs/papers/2405.16040v1.pdf`; zero occurrences of manifold/Riemannian/torus in the full text.
- [11] Matt Jacobs, Ekaterina Merkurjev, and Selim Esedoğlu. Auction dynamics: A volume constrained MBO scheme. *Journal of Computational Physics*, 354:288–310, 2018. AUTHORS CORRECTED 2026-08-21, verified against dblp (`journals/jcphy/JacobsME18`) and ScienceDirect. This entry previously read “Jacobs, Kim and Léger”, an error that originated in the archived A–E proposal (`docs/reference/phase1_highn_proposal_ABCDE.original.md`) and propagated into

seven documents before being checked. Volume, pages and year were correct throughout. Note Esedoglu is an author of both this and `esedoglu2015threshold`.

- [12] Jun Kitagawa, Quentin Mérigot, and Boris Thibert. Convergence of a Newton algorithm for semi-discrete optimal transport. *Journal of the European Mathematical Society*, 21(9):2603–2651, 2019.
- [13] Stuart P. Lloyd. Least squares quantization in PCM. *IEEE Transactions on Information Theory*, 28(2):129–137, 1982.
- [14] Quentin Mérigot. A multiscale approach to optimal transport. *Computer Graphics Forum*, 30(5):1583–1592, 2011.
- [15] Barry Merriman, James K. Bence, and Stanley J. Osher. Motion of multiple junctions: A level set approach. *Journal of Computational Physics*, 112(2):334–363, 1994.
- [16] Luciano Modica. The gradient theory of phase transitions and the minimal interface criterion. *Archive for Rational Mechanics and Analysis*, 98(2):123–142, 1987. Classical Γ -convergence result for the Modica–Mortola functional; source of the interface constant $c_W = 2 \int_0^1 \sqrt{W}$ used in the Phase 1 energy.
- [17] Luciano Modica and Stefano Mortola. Un esempio di γ -convergenza. *Bollettino dell’Unione Matematica Italiana B (5)*, 14:285–299, 1977. The original Γ -convergence example behind the Phase 1 relaxation energy.
- [18] Yves van Gennip, Nestor Guillen, Braxton Osting, and Andrea L. Bertozzi. Mean curvature, threshold dynamics, and phase field theory on finite graphs. *Milan Journal of Mathematics*, 82(1):3–65, 2014.